▸ GAI OBSERVE LLC · POC LEARNING LAB

Six control patterns for enterprise agentic AI

The Business Case — POCs 01–06

Scripted demonstrations on synthetic data. Every figure in this deck is ILLUSTRATIVE — drawn from runs recorded in the public repository, quoted with that label.

GAI-Engineering/poc-learning-lab · MIT (code) · CC BY 4.0 (content) · October 2026

$2.58B → $24.5B

Enterprise agentic AI market, 2024 → 2030 (Grand View Research), growing at a CAGR of 46.2% from 2025 to 2030. The spend is arriving whether the controls are ready or not.

40% of enterprise apps

will feature task-specific AI agents by the end of 2026 (Gartner, August 2025) — up from less than 5% in 2025.

>40% cancelled

of agentic AI projects will be cancelled by the end of 2027 (Gartner, June 2025) — escalating costs, unclear business value or inadequate risk controls.

$1.5B → $41.8B

Enterprise agentic AI software market, 2025 → 2030 (Omdia, September 2025) — a second forecaster's cut of the same surge; forecasts vary by definition and scope.

The gap

Growth and cancellation are the same market read twice. The six patterns in this lab are the controls that decide which side of it a deployment lands on.

Sources: grandviewresearch.com/industry-analysis/enterprise-agentic-ai-market-report · gartner.com/en/newsroom/press-releases/2025-08-26-gartner-predicts-40-percent-of-enterprise-apps-will-feature-task-specific-ai-agents-by-2026 · omdia.tech.informa.com/pr/2025/sep/new-omdia-analysis-shows-agentic-ai-outpacing-growth-rates-of-traditional-generative-ai

An agent that checks its own claims before anyone acts on them.

- 13 region claims · 10 survived verification · survival rate 0.769
- Flag rule: confidence < 0.60 = GUESS (flagged, not hidden)
- The 3 failed claims are exactly the 3 flagged (0.42 / 0.35 / 0.48) — in the demo this holds by construction; the lesson is the loop, not the rate
- Flip one verification outcome and the headline moves to 0.692 — try it live in the notebook (§ "Change one outcome")

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POC 01 · CTO CONCLUSIONS

An Agent That Sees

Grounded perception + a verification loop

KEY DATA POINTS · SCRIPTED DEMONSTRATION (ILLUSTRATIVE)

13 claims
region-grounded, each with a confidence

10 / 13 survived
survival rate 0.769 against ground truth

3 flagged = 3 failed
every GUESS (below 0.60) failed the check — by construction in the demo

0 hidden
low confidence is flagged, never buried

THE CTO CONCLUSION

Perception without a verification loop is unpriced risk.
The survival rate is the price tag — demand it before scale.

Run it as a measured pilot: your scenes, human checkers, one site.

Full notebook, code and evidence: github.com/GAI-Engineering/poc-learning-lab · MIT / CC BY 4.0 · Synthetic data only

The transcript is the system of record — coordination you can replay.

- 12 transcript messages · 4 tasks · 2 agents · 0 side channels
- 3 unasked-for findings: $78.00 unused gym charge (Feb–Mar) · coffee subscription billed 3x in March vs 2x · second streaming sub started March
- State = the transcript file; replay and audit read it alone

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POC 02 · CTO CONCLUSIONS

Conversation Is a Coordination Primitive

The transcript is the system of record

KEY DATA POINTS · SCRIPTED DEMONSTRATION (ILLUSTRATIVE)

12 messages
the entire collaboration — nothing outside the transcript

4 analysis tasks
coordinated by two agents with no shared memory

3 unasked-for findings
incl. a $78 unused charge the analyst never queried

100% replayable
audit = read the transcript; there is nowhere else for state to hide

THE CTO CONCLUSION

If agent work cannot be replayed from one record,
you do not have automation — you have anecdotes.

Run it as a measured pilot: one workflow, transcript-of-record on.

Full notebook, code and evidence: github.com/GAI-Engineering/poc-learning-lab · MIT / CC BY 4.0 · Synthetic data only

Four lines of contract, linted before the agent ever runs.

- Contract = 4 lines (Role · Output · Do-Nots · Signal)
- Good contract: lint CLEAN · Bad contract: 3 flags (MISSING Do-Nots, WEAK_SIGNAL, VAGUE_OUTPUT)
- Extraction: 4 obligations · termination on the typed signal EXTRACTION COMPLETE (check returned SIGNAL, never a prose word)

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POC 03 · CTO CONCLUSIONS

RODS — Contract Before Capability

Four lines, linted before the agent runs

KEY DATA POINTS · SCRIPTED DEMONSTRATION (ILLUSTRATIVE)

4 lines

Role · Output · Do-Nots · Signal — the whole contract

3 flags

on the bad contract: missing Do-Nots, vague output, weak signal

0 flags

on the good contract — it runs; the other never executes

1 typed signal

EXTRACTION COMPLETE ends the run — never a prose word

THE CTO CONCLUSION

Contract failures are preventable before runtime.
Four lines and a checker price the risk before deployment.

Adopt it as the intake gate: no contract, no run.

Full notebook, code and evidence: github.com/GAI-Engineering/poc-learning-lab · MIT / CC BY 4.0 · Synthetic data only

Every outcome is labelled, and the label teaches the next attempt.

- 5 attempts · 1 ERROR · 1 EMPTY · 3 SUCCESS — outcomes computed from the run record (runs/attempts.csv), never typed into the summary
- The error was a wrong key ('amt'); the recovery habit — print the schema first — is attempt 2, and the revised query succeeds
- EMPTY means the question was wrong (category 'gym' matches no rows); the exact name from the totals ('gym-unused') succeeds
- Finding: gym-unused totals $78.00 (2 x $39.00, 2026-02 and 2026-03), traced to rows in data/transactions.csv (22 rows)
- Change a figure in the CSV and the finding moves — the perturbation test moves the gym amounts and watches $78.00 become $98.00 (tests/test_perturbation.py)

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POC 04 · CTO CONCLUSIONS

The Error Is the Curriculum

Five attempts, typed outcomes, one finding traced to the rows

KEY DATA POINTS · SCRIPTED DEMONSTRATION (ILLUSTRATIVE)

5 attempts

every one typed — 1 ERROR · 1 EMPTY · 3 SUCCESS, from the run record

2 failure kinds

ERROR = the code was wrong · EMPTY = the question was wrong — labelled apart

$78.00 found

an unused subscription, 2 x $39.00, traced to the actual rows

1 recovery habit

print the schema first — the error became the next attempt's input

THE CTO CONCLUSION

An agent's failures are its training data.
Price the loop by how it recovers, not by its first try.

Adopt it where answers must explain themselves: type every failure.

Full notebook, code and evidence: github.com/GAI-Engineering/poc-learning-lab · MIT / CC BY 4.0 · Synthetic data only

Who speaks next is a design decision — count what each shape costs in calls.

- Same job, three shapes, counted from runs/routing-comparison.csv: round robin 6 agent turns / 6 total calls · selector 3 agent turns + 4 routing calls / 7 total · swarm 4 agent turns + 3 hand-offs / 4 total
- All three modes prepared the same pack (specialty orthopaedics, 2 slots — Tue 09:30, Wed 14:10 — unbooked, awaiting human confirmation)
- These counts are the mechanics of this script — round robin runs its sequence twice by construction; the swarm takes a verification hand-back by construction. They are not live routing costs.
- A router that can never finish is capped and ends INCOMPLETE (tests/test_perturbation.py, failed-router case)



Structured disagreement allocates scarce human review to the closest fights.

- 3 positions debated (runs/verdicts.csv): C1 auto-renewal at +8% — REJECT (strength 5 for / 11 against) · C2 24-hour incident-notice duty — PROCEED (8 / 5) · C3 uncapped liability — REJECT (2 / 14)
- Closest fight: C2, balance 0.62 — the most balanced argument in the script and the reviewer's first stop
- The balance map is an illustrative argument-balance prompt for human review: it locates no genuine uncertainty, is not a risk register, and a real position requires independent evidence and counsel review
- Adding a fourth position produces four verdicts — the counts are computed, not typed (tests/test_perturbation.py)

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POC 06 · CTO CONCLUSIONS

Disagreement Is a Sensor

Structured debate that rations human review — and decides nothing

KEY DATA POINTS · SCRIPTED DEMONSTRATION (ILLUSTRATIVE)

3 positions
contract clauses debated from the same evidence, both ways

2 REJECT · 1 PROCEED
verdicts computed from the strength of each side's case

0.62
the closest fight — the 24-hour notice duty; the reviewer's first stop

0 decisions
made by the script: it allocates review; humans and counsel decide

THE CTO CONCLUSION

Review time is the scarcest control in governance.
Spend it where the argument is hardest to call.

Adopt it as the review queue: both cases on record, closest first.

Full notebook, code and evidence: github.com/GAI-Engineering/poc-learning-lab · MIT / CC BY 4.0 · Synthetic data only

THE PORTFOLIO VIEW — WHERE THE NEXT EFFORT GOES

Invest first — POC 03 (RODS)

The contract pattern is the most transferable asset in the lab: four lines, a linter, a typed completion signal. It generalises to every agent the lab will ever ship.

Gated — POC 02 (transcripts)

The system-of-record pattern proceeds behind its durability gates: persistence, replay and failure recovery proven before any stronger claim.

Teaching demo — POC 01 (perception)

Held as a teaching artefact until a real perception adapter replaces the scripted scene. Its lesson — the verification loop — is already load-bearing in POC 04.

Day 2 (POCs 04–06)

The analyst–executor loop, group-chat routing shapes and structured debate extend the same discipline: labelled outcomes, counted calls, disagreement treated as a sensor.

METHOD — HOW THIS LAB EARNS TRUST

- Every result is computed from data files in the repo — change an input and the result moves; the Day 2 notebooks (04–06) carry perturbation tests that prove it, and Day 1’s carry change-one-outcome exercises.
- Modes are printed on the face of the work: INTERIM notebooks are scripted demonstrations; a FINAL promotion requires a real recorded run, reproducible end to end, and passes the evaluator’s full QA.
- Releases are signed by an independent evaluator of record, with conditions recorded in RELEASES.md and closed only on evidence.
- Figures of record are the data-drawn SVG infographics in business/infographics/; generated images are checked figure-by-figure at full size before they are used.
- Corrections are made in public, in the commit history — including this deck’s predecessors.